Rotation of the axis of rotation

Euler's Theorem states that any orientation $R \in SO(3)$ is equivalent to a rotation about a fixed axis $\omega \in \mathbb{R}^3$ through an angle $\theta \in [0, 2\pi)$ (MLS p. 31). In this statement ω is a unit vector. What happens if this vector ω is rotated through the rotation defined by R? The resulting vector must be the same as the original vector ω. This code, however, also proves this, by showing that

$\omega = e^{\hat{\omega}\theta}\omega$

```
clearvars, clc

syms omega1 omega2 omega3 theta real

omega = [omega1; omega2; omega3]
```

omega =

$$\begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$

```
Rab   = rodrigues(omega,theta)
```

Rab =

$$\begin{pmatrix} (\cos(\theta) - 1)\ \left(\omega_2{}^2 + \omega_3{}^2\right) + 1 & -\omega_3 \sin(\theta) - \sigma_3 & \omega_2 \sin(\theta) - \sigma_2 \\ \omega_3 \sin(\theta) - \sigma_3 & (\cos(\theta) - 1)\ \left(\omega_1{}^2 + \omega_3{}^2\right) + 1 & -\omega_1 \sin(\theta) - \sigma_1 \\ -\omega_2 \sin(\theta) - \sigma_2 & \omega_1 \sin(\theta) - \sigma_1 & (\cos(\theta) - 1)\ \left(\omega_1{}^2 + \omega_2{}^2\right) + 1 \end{pmatrix}$$

where

$$\sigma_1 = \omega_2\, \omega_3\ (\cos(\theta) - 1)$$

$$\sigma_2 = \omega_1\, \omega_3\ (\cos(\theta) - 1)$$

$$\sigma_3 = \omega_1\, \omega_2\ (\cos(\theta) - 1)$$

```
resulting_vector = expand(Rab*omega)
```

resulting_vector =

$$\begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$